PRINCE CHAUHAN

Senior Quantitative Researcher & Portfolio Manager | Systematic Derivatives & Volatility Arbitrage

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SUMMARY

Quantitative Researcher and Portfolio Manager with nine years of systematic derivatives and volatility arbitrage experience across Indian index options (Nifty, Bank Nifty, Sensex) and US derivatives (CBOE SPX, CME). Architected an institutional dual-market quantitative trading desk operating ~INR 100 Cr notional capacity with sub-millisecond execution dispatch, point-in-time trade ledgers, dynamic Greeks hedging, and automated pre-trade risk controls. Chartered Accountancy background (CA Final, Group 1 cleared, 78%) ensures rigorous econometric overfitting defense, zero lookahead bias, and full transaction cost accounting on every traded leg.

EXPERIENCE

SMC Global Securities Ltd. - Senior Quantitative Analyst & Lead Strategist — Dec 2024 - Present

- Architected and operate an institutional systematic options desk managing ~INR 100 Cr notional capacity across NSE index options and multi-asset baskets under strict SPAN margin sizing and automated daily loss stops.
- Engineered Indian Trading Desk Quants Nerve: low-latency Python execution platform featuring asyncio WebSocket tick listeners, memory-mapped IPC, real-time Greeks risk engine, and sub-millisecond order routing via DhanHQ API.
- Engineered USA Desk Quants Nervo: extended desk to US markets streaming Databento OPRA Level 3 feeds and CME futures (ES, NQ); built systematic SPX 0DTE options execution bots and real-time Net Dealer Gamma (GEX) tracking.
- Operationalize thirteen systematic strategy families in live production forward execution with leg-by-leg trade ledgers: delta-hedged straddles, opening-range option buying/selling, IV z-skew arbitrage, and Bank Nifty stat-arb baskets.
- Implemented econometric validation gate: walk-forward testing, parameter-plateau checks, 2x cost stress, Probabilistic and Deflated Sharpe ratios, Probability of Backtest Overfitting (PBO via CSCV), and Monte Carlo drawdowns.
- Engineered pre-trade RMS: dynamic Greeks limits, India VIX volatility circuit breaker, automated exchange freeze-limit slicing (1,800 Nifty / 900 Bank Nifty), 15:20 IST forced exit escalator, and token bucket rate limiters.
- Formulated transaction cost analysis (TCA) model: dynamic bid-ask spread widening, square-root market impact, order book imbalance (OBI) liquidity penalties, and full Indian statutory charge schedules.

Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research — Mar 2017 - Nov 2024

- Managed systematic multi-leg options volatility books (iron condors, calendars, butterflies) on Nifty and Bank Nifty, executing automated intraday delta, gamma, and vega rebalancing protocols.
- Engineered volatility surface engine: Black-Scholes IV inversion (Newton-Raphson with Brent fallback), SVI smile fitting with Gatheral butterfly and calendar no-arbitrage checks, and dealer gamma (GEX) profiles.
- Developed unsupervised machine learning regime classifier (k-means clustering on Parkinson realized volatility, IV/RV ratio, and rolling skew) with strict 1-bar lag, reducing tail drawdown by 28%.
- Built limit order queue position simulator and adverse selection model saving 1.8 bps in crossing costs.

Indian Oil Corporation Ltd.; CA Matta & Associates - Treasury & Financial Modeling; Audit & Tax — 2014 - 2017

- Conducted corporate treasury cash-flow forecasting and MIRS financial modeling at IOCL (2016-2017); performed statutory audit and corporate valuation at CA Matta & Associates (2014-2016).

DESK RESEARCH & BACKTESTING

- Audited multi-strategy portfolio backtest (Jan 2021 to Sep 2026, 1,411 sessions of 1-minute data, 0.50% cost on every leg): achieved 24.6% annualized return after costs, Sharpe 3.67, profit factor 2.09, and worst drawdown under 4%.
- Maintained institutional backtest library of 13 option strategies across 1-minute NSE/BSE and US index data. Automated delivery of 4 canonical deliverables: executive PDF tear sheet, VIX-regime attribution, Excel book, and CSV ledger.
- Engineered data pipeline integrity gates: tick-to-Parquet reconciliation, coverage audits for zero missing minutes, corporate actions adjustments, and point-in-time trade execution verification.
- Researched alpha signals feeding the book: Bank Nifty Johansen cointegration basket stat-arb, IV skew z-scores across strikes, order book imbalance (OBI), VPIN order-flow toxicity, and variance risk premium (VRP) harvest.
- Formulated risk and regime models: k-means market clustering, Monte Carlo dynamic VaR, and genetic parameter optimization scored strictly on Deflated Sharpe to eliminate selection bias.

OPEN-SOURCE RESEARCH CODE

github.com/prince-chauhan-pc/quant-research-portfolio - tested Python modules with automated unit tests and CI. Synthetic seeded data only; proprietary execution logic excluded.

- Point-in-time backtester with SPAN-margin lot sizing, per-leg friction drag, Sharpe / Sortino / Calmar and trade ledger.
- Volatility surface model: Black-Scholes IV inversion, SVI smile fit with butterfly/calendar arbitrage checks, and GEX profiles.
- Pairs and basket cointegration (Johansen trace test, Engle-Granger ADF, half-life, z-scores), order queue simulator, and DSR gate.

SKILLS

- **Quantitative Rigor:** Deflated & Probabilistic Sharpe, CSCV overfitting tests, walk-forward analysis, Johansen cointegration, time series, Monte Carlo VaR, variance risk premium, statistical arbitrage.
- **Derivatives & Greeks:** Options pricing, Greeks (Delta, Gamma, Vega, Theta, Vanna, Volga), SVI volatility surfaces, Net Dealer Gamma (GEX), volatility skew, straddles, iron condors, SPAN margin optimization.
- **Microstructure & ML:** Limit order books (L2/L3), order book imbalance (OBI), bid-ask spread TCA, market impact, VPIN, adverse selection, machine learning clustering (k-means), feature engineering, PCA, regularization.
- **Systems & Latency:** Python 3.12 (Polars, NumPy, SciPy, pandas, scikit-learn), asyncio, WebSockets, memory-mapped IPC, concurrency, low-latency dispatch, Flask, SQLite, DuckDB, Parquet, pytest, Git CI/CD.
- **Markets & Gateways:** India NSE/BSE F&O (DhanHQ API v2); US CBOE SPX & CME Futures (Databento Level 3 OPRA, Alpaca).

EDUCATION & REGISTRATION

- **B.Com (Honours):** Kirori Mal College, University of Delhi, 2011 - 2014. First Division (71%).
- **CA Final (Group 1 cleared, 78%):** Institute of Chartered Accountants of India, 2018.
- **Registration:** SEBI Registered Research Analyst (NISM Series-XV).